

Empirical Proof Record: VALIDATED

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Proof ID: b09222ef9dc216df233ca182eb10a95a...

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1. Claim

Across 30 synthetic (x, y) pairs of size 100 (seed=20260518), “scipy.stats.spearmanr” and “pingouin.corr(method='spearmanr)’” produce Spearman ρ values that agree to $\leq 1e-09$. (RFC 0002 Phase E1: cross-framework validation at the statistical-primitive layer.)

- **Operationalisation:** Generate N (x, y) pairs where $x \sim \text{Normal}(0, 1)$ and $y = \rho_{\text{target}} \times x + \sqrt{1 - \rho_{\text{target}}^2} \times \text{Normal}(0, 1)$ for a sweep of ρ_{target} . Compute Spearman ρ under both backends. Compute $\text{max_abs_diff} = \max \text{ over pairs of } |\rho_{\text{scipy}} - \rho_{\text{pingouin}}|$. VALIDATED iff $\text{max_abs_diff} \leq \text{tolerance}$.
- **Threshold:** `max_absolute_spearman_difference <= 1e-09 correlation`
- **H0:** scipy and pingouin disagree on Spearman ρ by more than the floating-point tolerance — at least one library has a defect or default-behaviour drift.
- **H1:** Both libraries compute identical Spearman ρ to floating-point tolerance across every pair.

2. Verdict

VALIDATED — observed 0.0

30 pairs checked; $\max |\rho_{\text{scipy}} - \rho_{\text{pingouin}}| = 0.000e+00 (\leq 1e-09 \text{ tol})$

3. Evidence

Pillar	Statistic	Value	95% CI	Library
spearman_scipy_vs_pingouin	max_absolute_spearman_difference	0.0	—	scipy 1.17.1

4. Pre-registration

- Registered at: 2026-05-23T23:12:37.867655+00:00
- Config hash: 4e41be096c5011fb2c2bed2d180c541c...
- Data hash: 4e41be096c5011fb2c2bed2d180c541c...

5. Signature

300de6fbf33951e7852f6814d33c650b4b49fc2f68c712e9e8f79063c16748a1